Lecture 1: State Space Models

Logistics

- Time: Tuesday and Thursday 2:00pm - 3:20pm
- Location: NSH 3002
- Instructor: **Changliu Liu** (cliu6@andrew.cmu.edu)
- Instructor office hours: Tuesday 3:30-4:30pm NSH 4525
- Teaching Assistant: **Yifan Sun** (yifansu2@andrew.cmu.edu)
- TA office hours: Tuesday 10:00-11:00pm NSH 4513
- Canvas: https://canvas.cmu.edu/courses/55422
- Piazza: https://piazza.com/class/mspzbk04s6kmf/
  - enroll through this link: https://piazza.com/cmu/fall2026/16714
- Repository: https://github.com/intelligent-control-lab/16-714

Course Topics

- Topics to be covered:
  - Optimal control
  - Model predictive control
  - Iterative learning control
  - Kalman filter, EKF, UKF
  - Model reference adaptive control
  - Policy gradient
  - Lyapunov methods
  - Constrained control
  - Koopman theory

Prerequisites

- Required:
  - Linear algebra
  - Linear systems and control
- Preferred
  - Probability theory
  - Statistical learning
  - Kinematics, Dynamics, and Control
- Prototyping skills: Matlab, **Python**, C, C++
- This course will be math heavy.
- Other related courses: Optimal Control and Reinforcement Learning (16-745), Statistical Techniques in Robotics (16-831), Nonlinear Control (18-776)

Assessment Structure

Homeworks	90%
Quiz	10%

Policies

- See course syllabus.
- Homework:
  - Starting 8/28, one homework every two weeks;
  - Expected workload: 6 hours per homework; should not take more than 12 hours.
  - **No late days** since Canvas will automatically release the solution after the due day.

Use of AI Tools

AI tools (e.g., ChatGPT, Gemini, Cursor) may be used to support and enhance your learning. However, their use must follow clear guidelines.

Prohibited Uses (considered academic dishonesty):

- Submitting AI-generated solutions for homework.
- Submitting AI-generated code for assignments.
- Submitting AI-generated solutions for quiz.

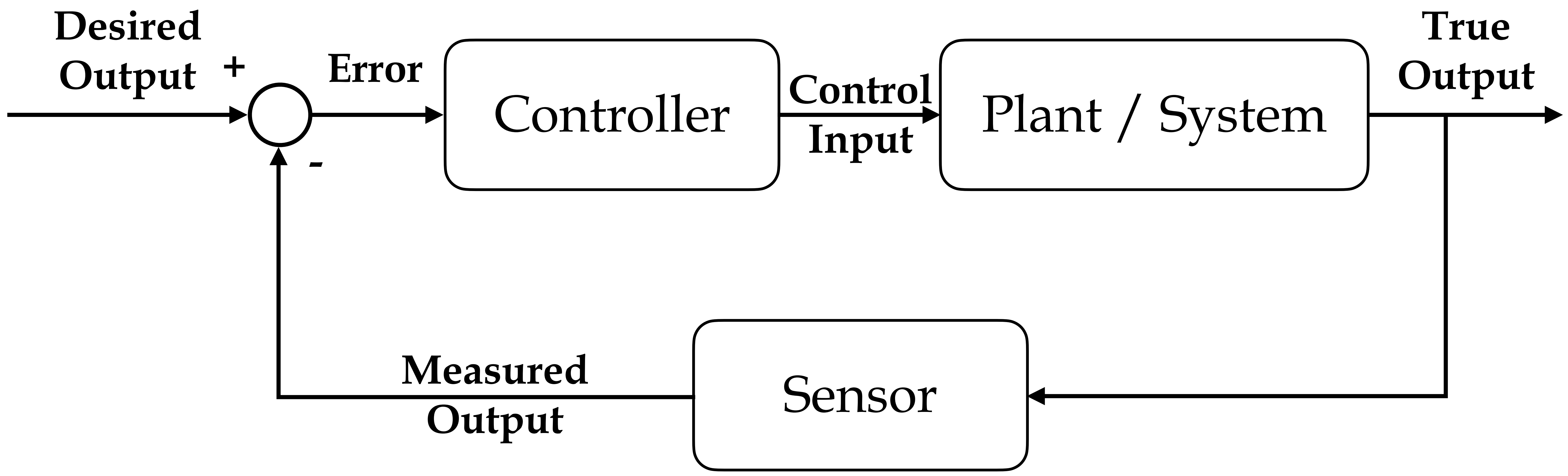
Permitted Uses (to enhance learning):

- Summarizing or reorganizing your lecture notes.
- Obtaining additional explanations of class topics.
- Clarifying or better understanding provided code.

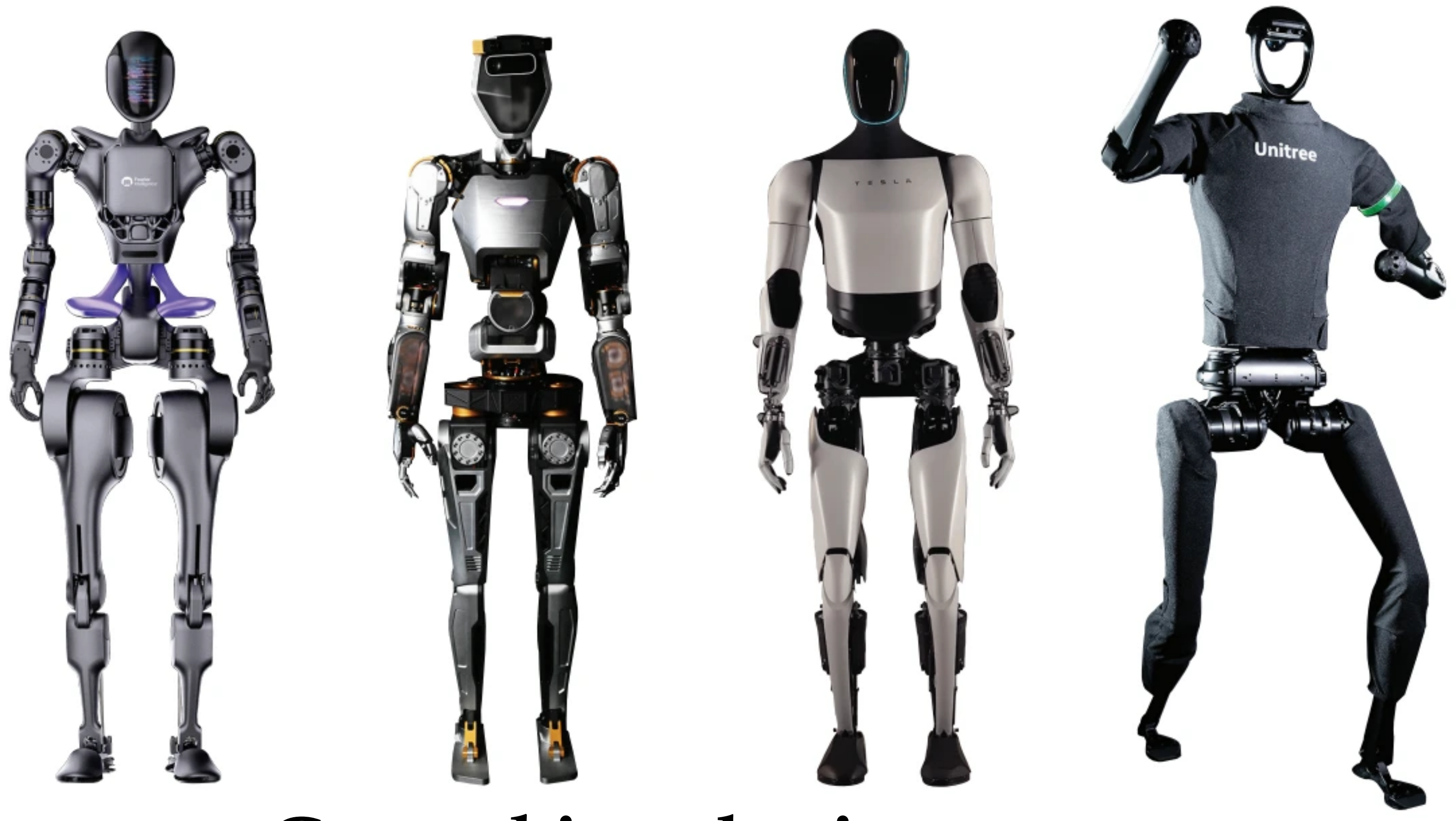
In short, you may use AI to **learn**, but not to **replace your own work**.

Questions?

What is Control Theory?



What is Control Theory?

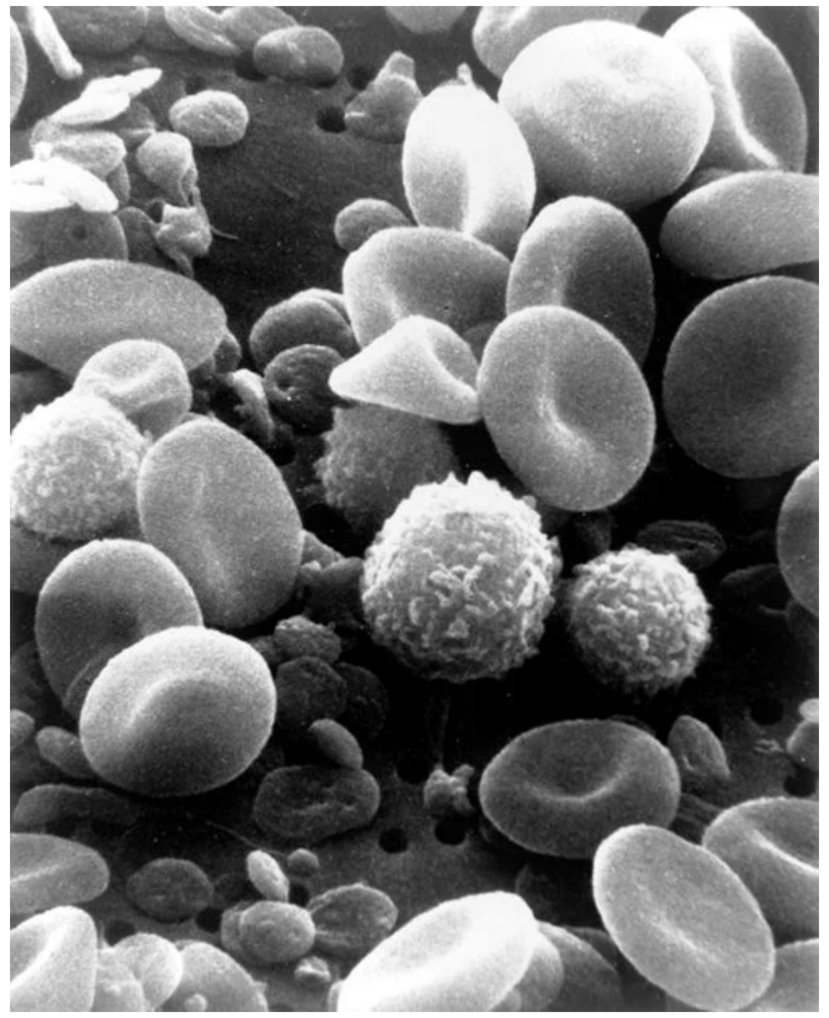


Control in robotics systems

Plant / System

Control in economic systems

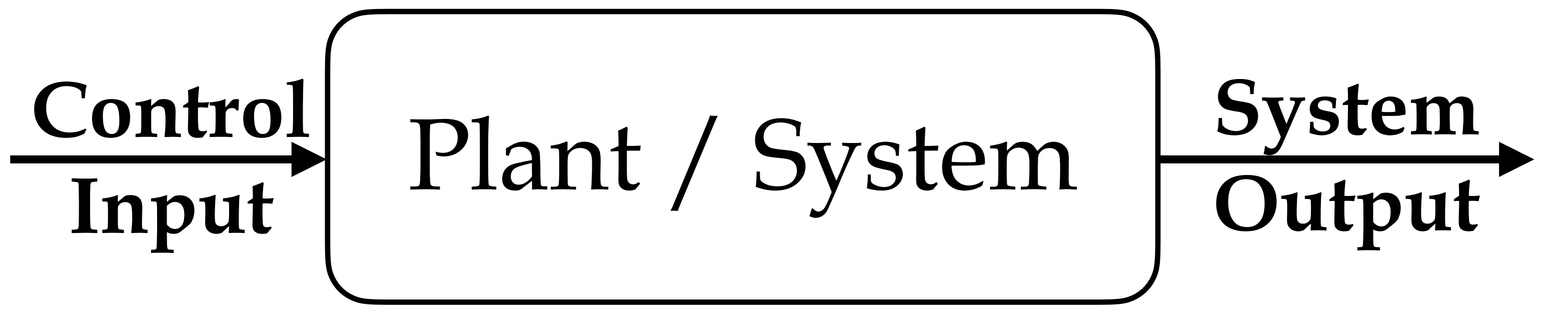
Control in chemical systems



Control in biological systems

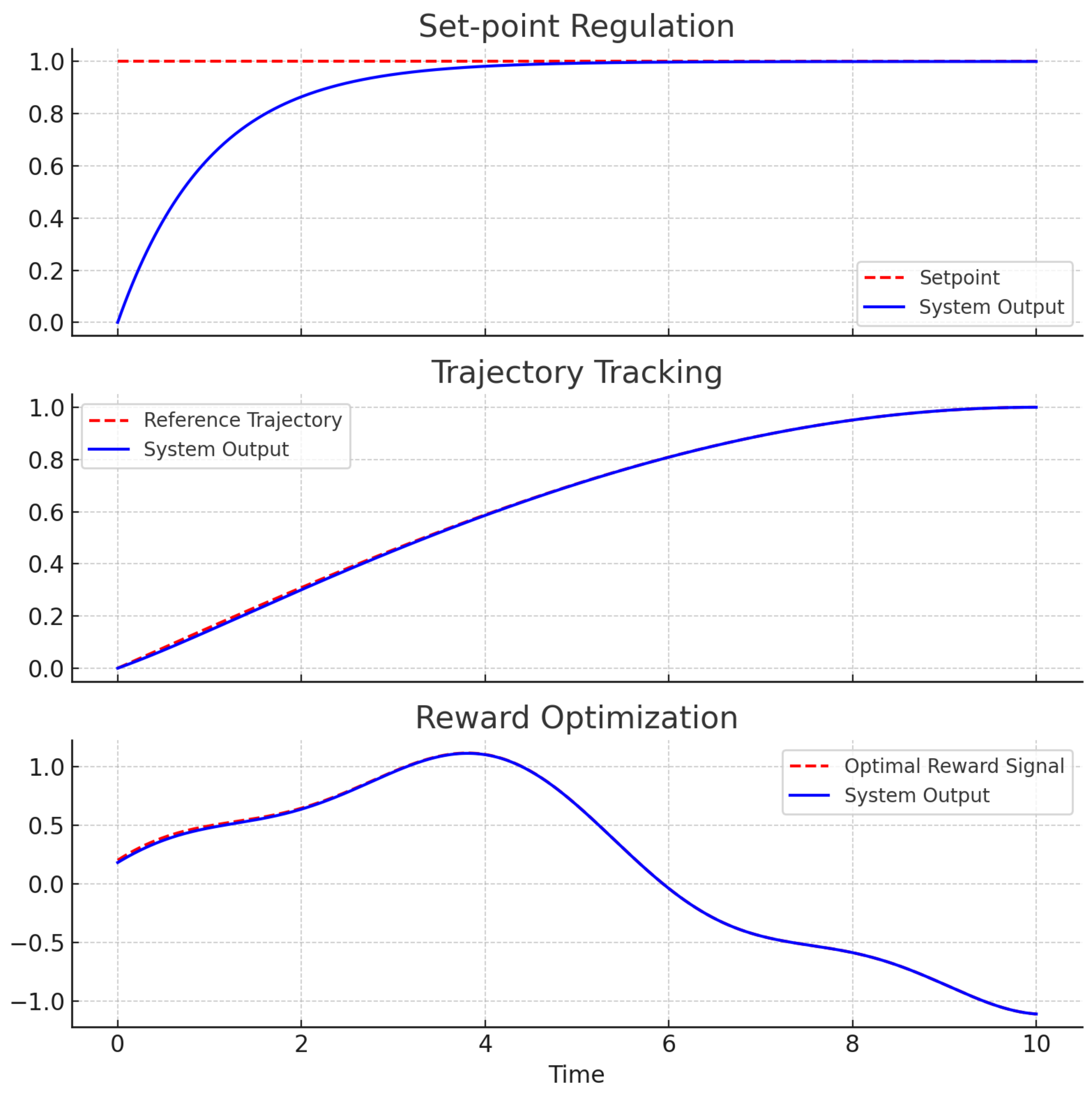
What is Control Theory?

Goal: Execute some control input such that the system output meets certain **requirements**



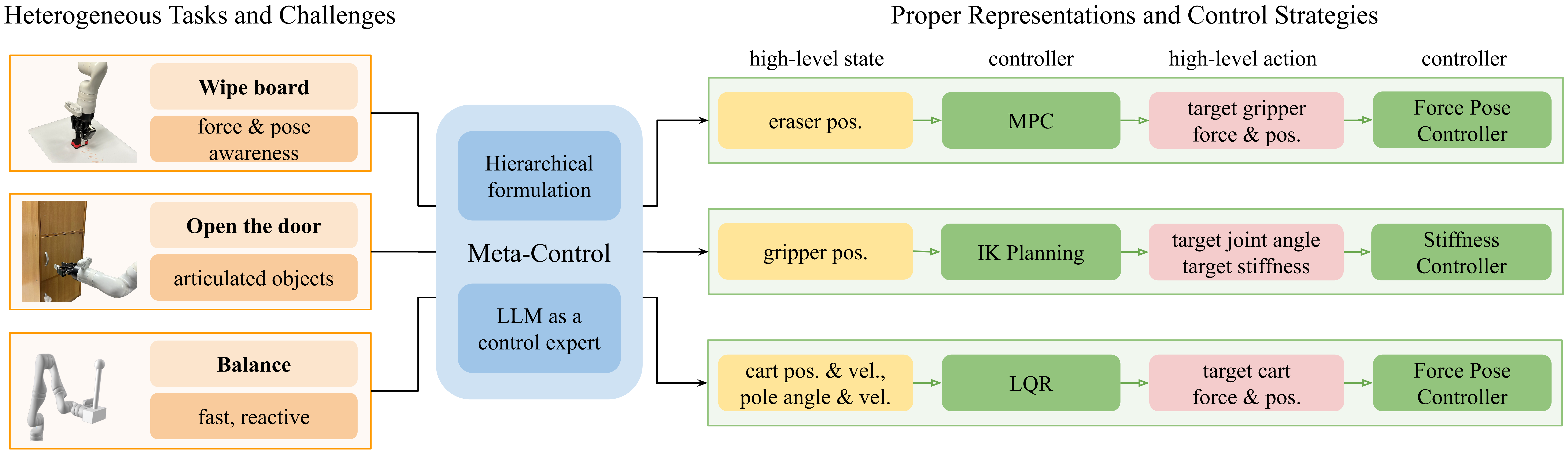
Requirements Studied in Control Theory

- **Set-point regulation**
  - a robotic arm keeps its welding torch fixed at the seam.
- **Trajectory tracking**
  - a drone follows a person's path.
- **Reward optimization**
  - an autonomous car chooses a merging path that balances safety and efficiency.



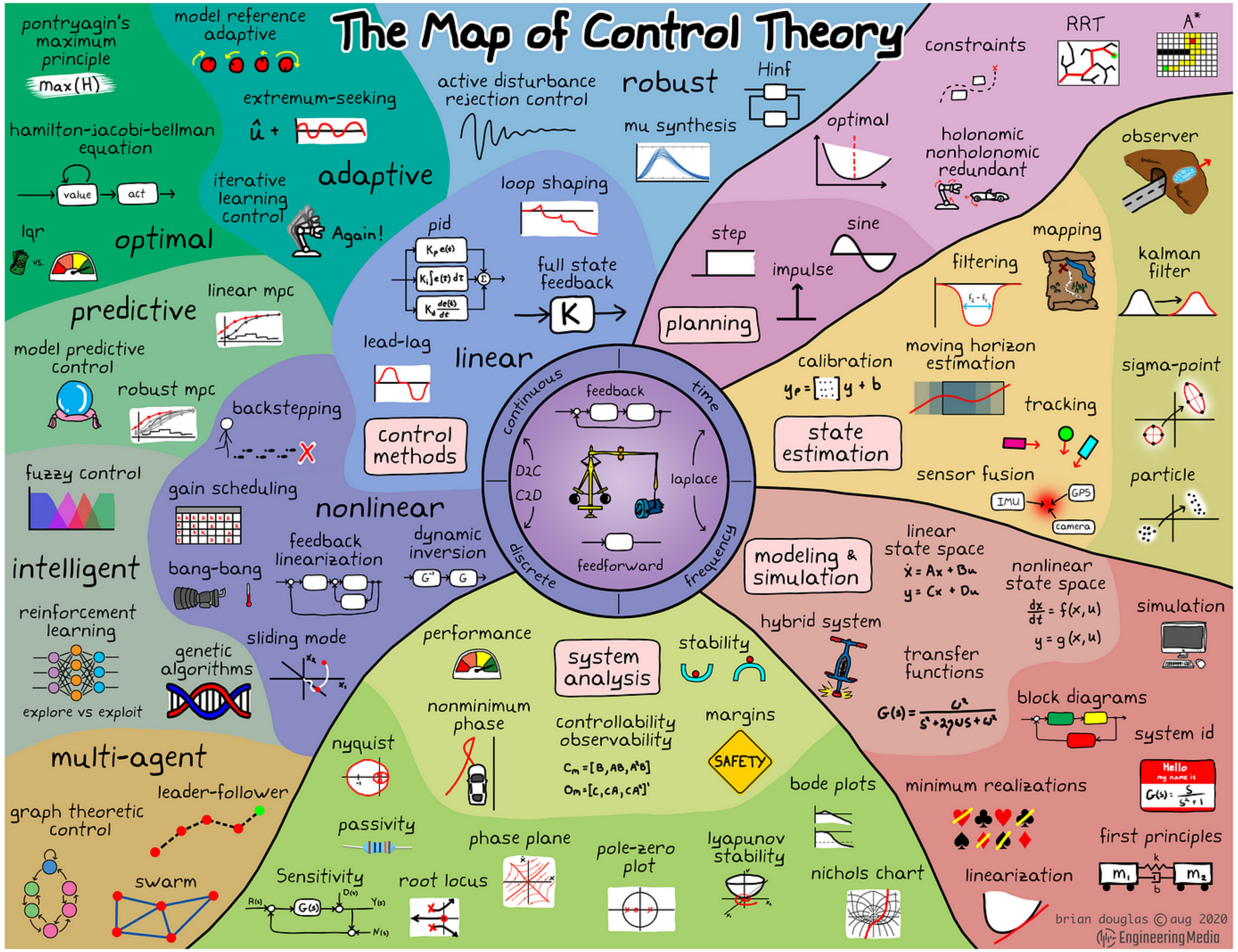
Can Single Control Strategy Work for All?

- Unfortunately NO
  - Requirements are diverse
  - System properties vary a lot



Meta-Control: Automatic Model-based Control Synthesis for Heterogeneous Robot Skills

The Map of Control Theory



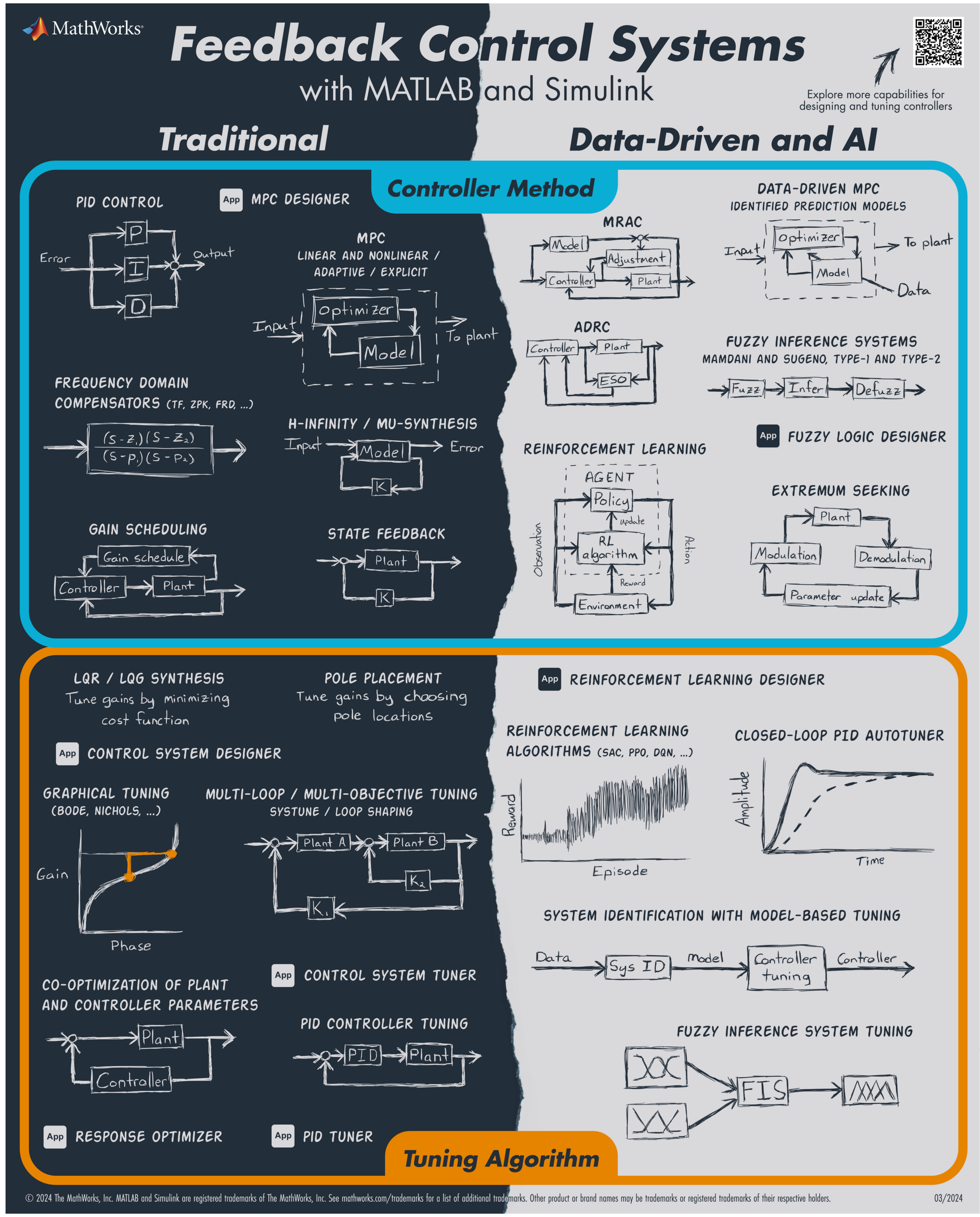
https://engineeringmedia.com/

Control vs Learning?

- Both are solving “control problems” or “decision making problems”
- Core difference:
  - Model-based approaches (control): need human control experts (or AI agents?) to digest the requirements, analyze the system property, then come up with controller design
  - Data-driven approaches (learning): need human learning experts (or AI agents?) to set up the training pipeline (e.g., how to collect data, how to set hyperparameters of the learning algorithm, etc)
- These result in different scalability, soundness, interpretability, etc.

https://medium.com/geekculture/control-theory-in-the-modern-era-is-it-dead-7aebe840a3f

Control vs Learning?



History of Control Theory

To be studied in this class

Classical Control

- Frequency domain-based tools
- Mainly useful for single-input-single-output (SISO) systems
- Still one of the main tools for the practitioning engineer

1940

Modern Control

- State-space approach to linear control theory
- Works for SISO and multi-input-multi-output (MIMO) systems
- Consider more diverse performance and robustness metrics

1960

Nonlinear Control

- Linearization methods
- Nonlinear optimization methods
- Various specialized techniques for control of nonlinear systems

1990

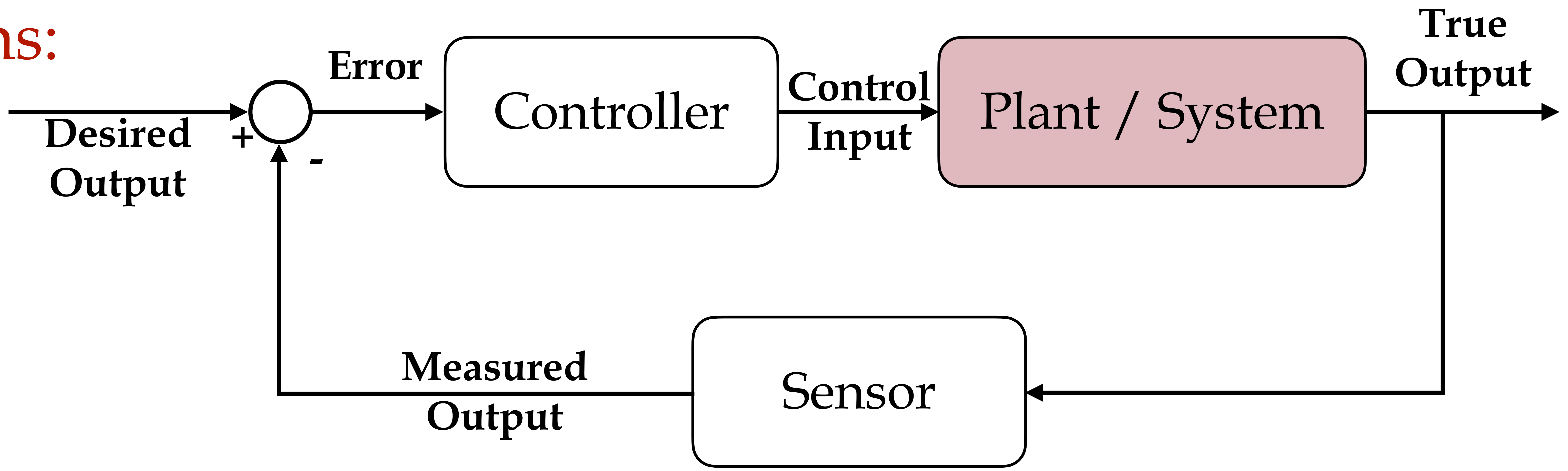
Now?

- Model-based + data-driven approaches
- Neural symbolic approaches
- ....

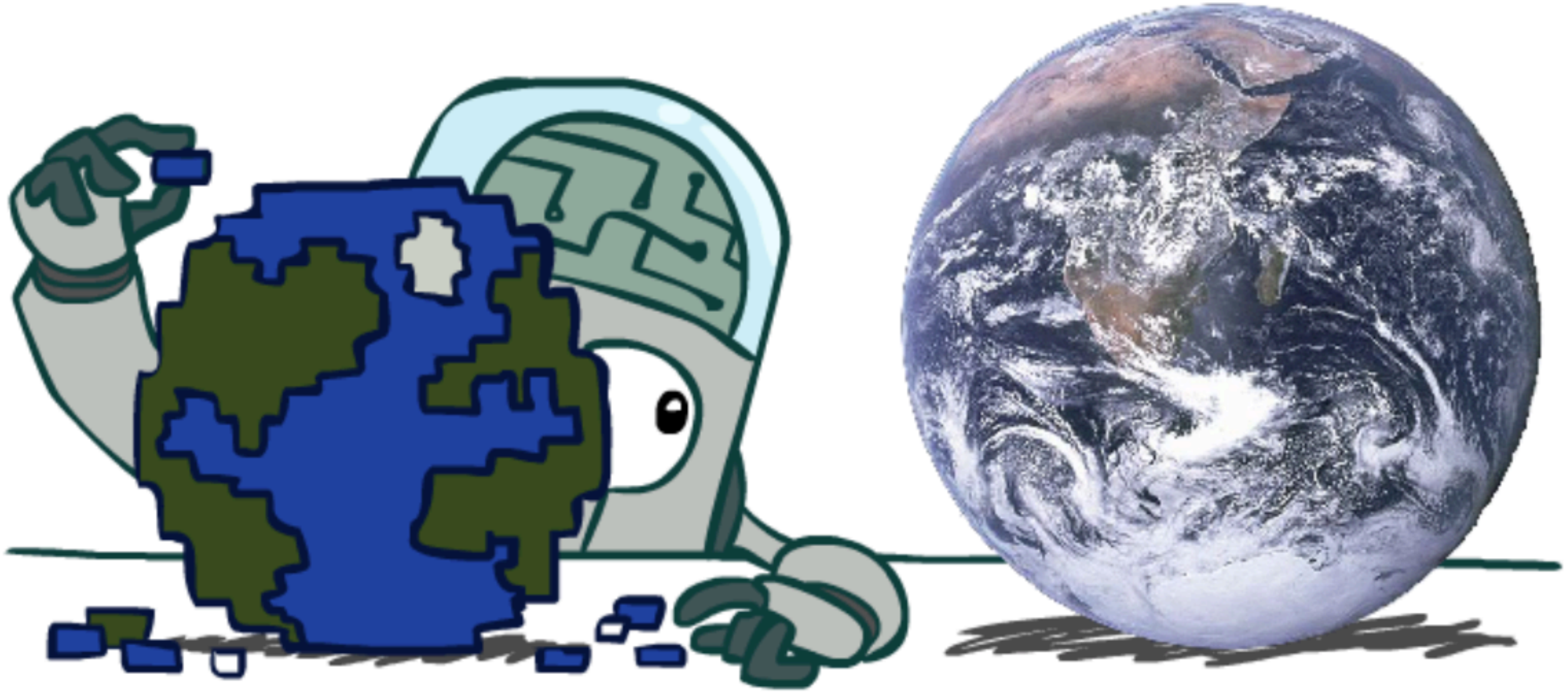
2010

This Lecture

- Describing dynamic systems:
  - State-space model
  - Markov decision process



Modeling



All models are wrong, but some are useful.

State Space

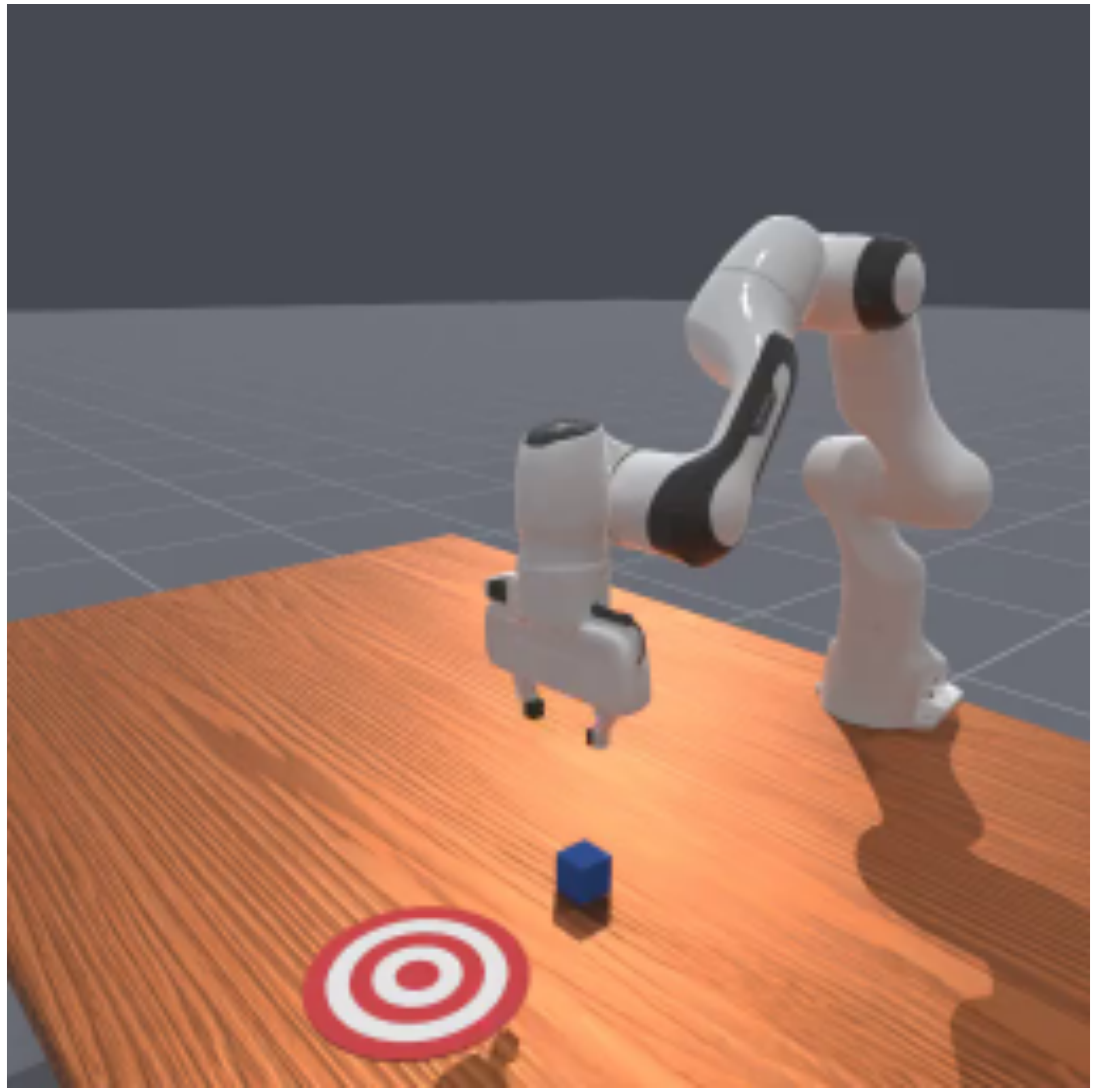
- The internal **state variables x** are:
  - the smallest possible subset of system variables
  - that can represent the entire state of the system at any given time
  - such that the future evolvement of the system only depends on the current state, not the past (Markov).

Markov

- What does Markov mean?
  - “Markov” generally means that given the **present** state, the future and the past are independent
- Choice of states is important

Andrey Markov
(1856-1922)

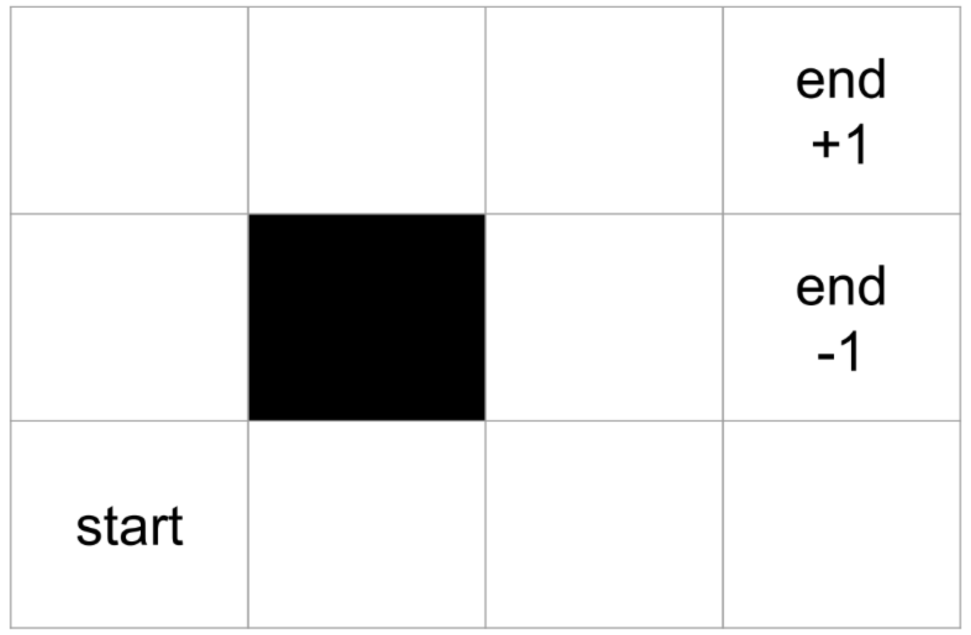
Example: What is the State?

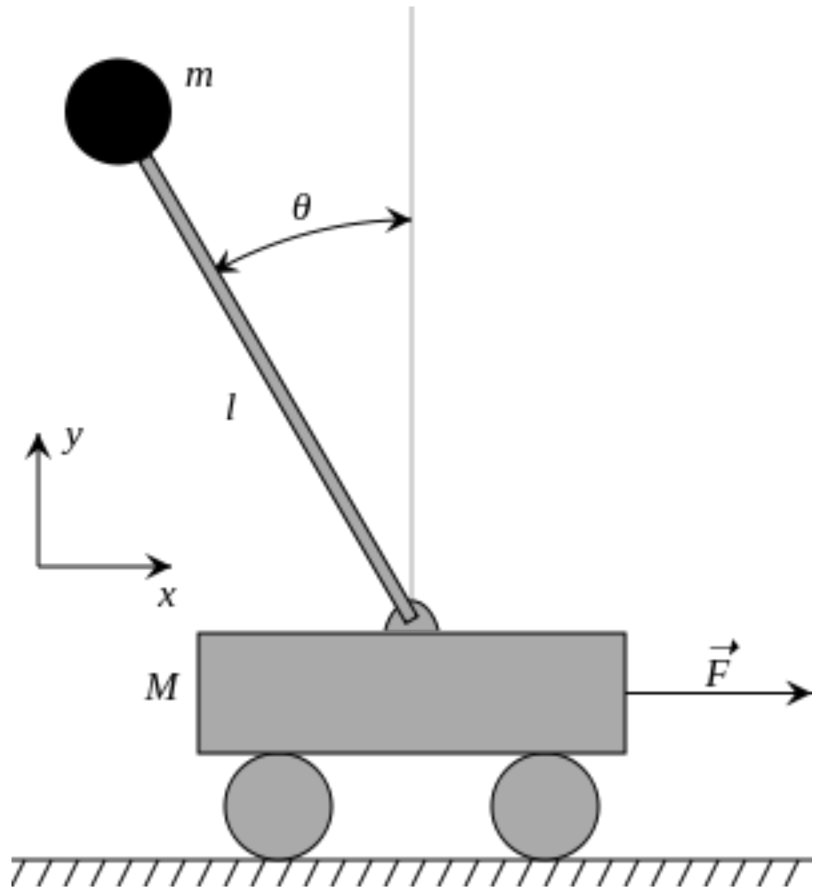


https://maniskill.readthedocs.io/en/latest/tasks/table_top_gripper/

State Space

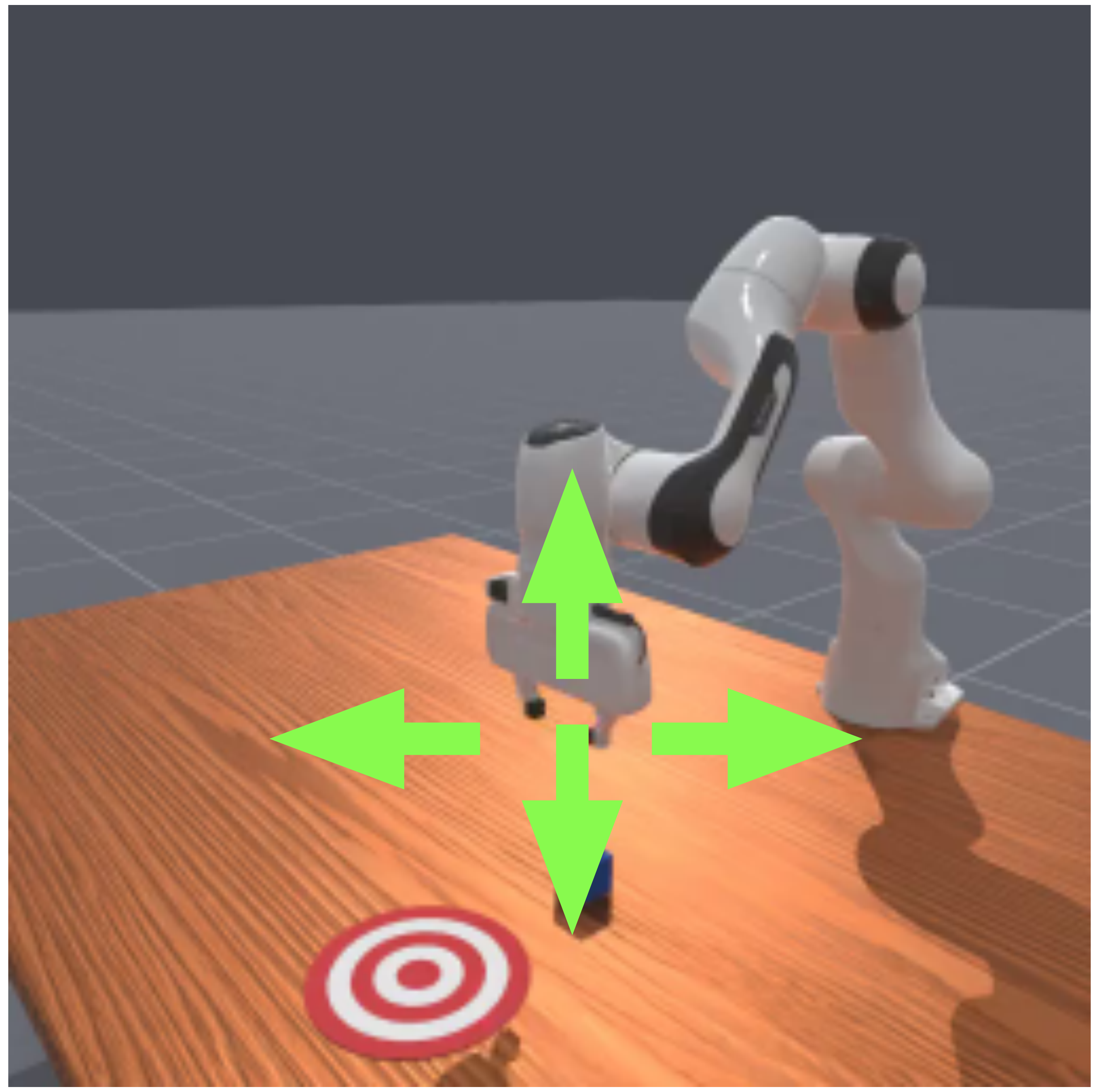
- **State space X** is the collection of all possible values of the state variables.
- **State values** refer to the value of the state variables.
- State values can be continuous or discrete
  - Example: grid world
  - Example: inverted pendulum



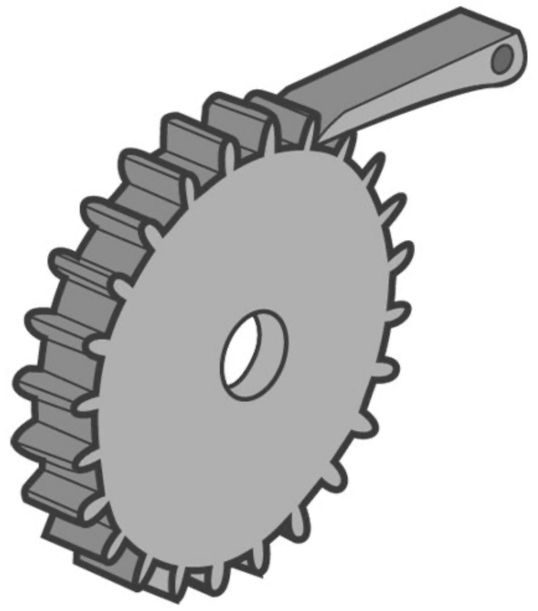


Control Space

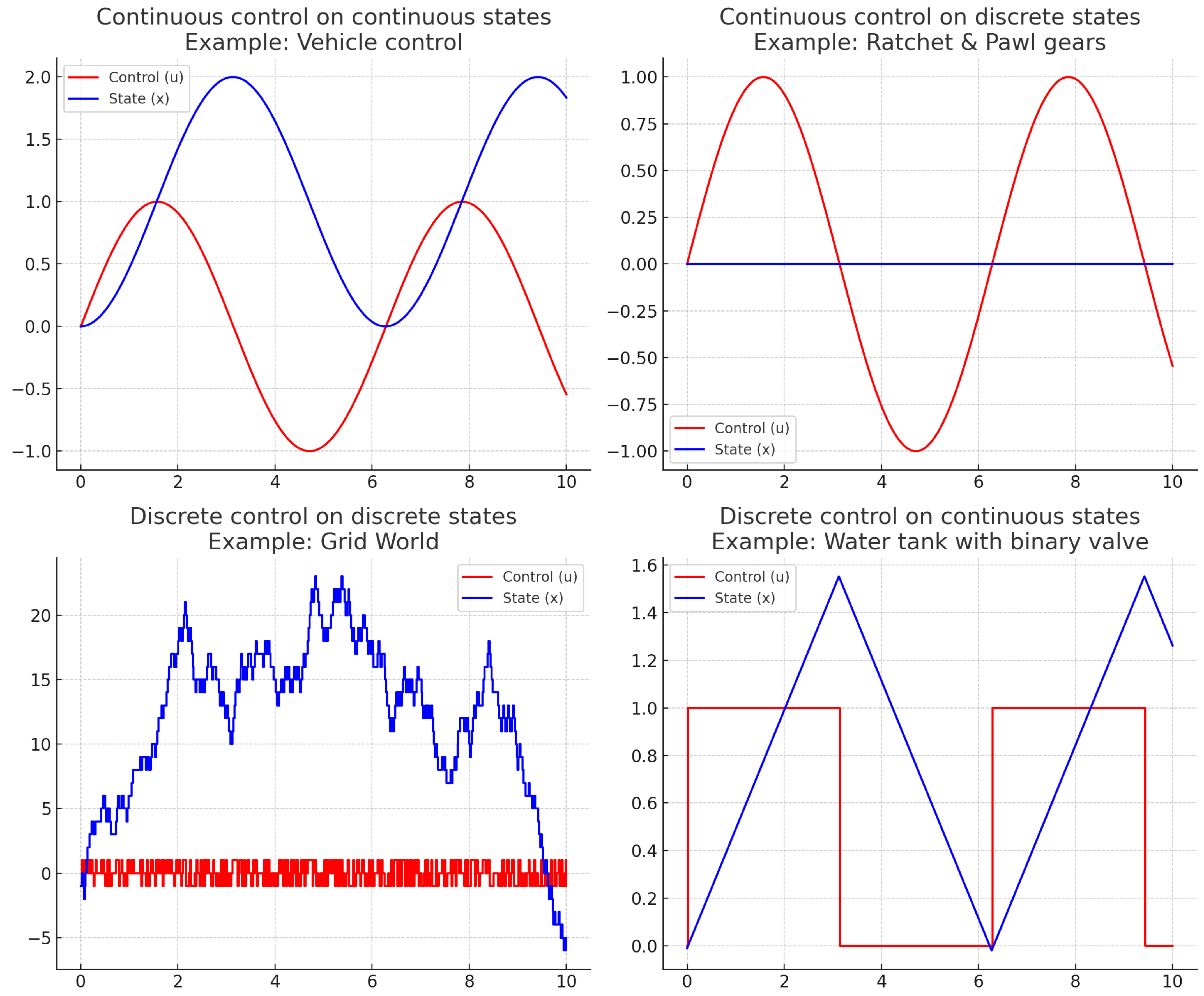
- Control space: the set of variables that the agent can choose to affect future states.
- Control signal - u; control space - U.
- The control signals can be continuous or discrete:
  - Example: lottery
  - Example: vehicle steering



Control Space



- It is possible to have:
  - Continuous controls on continuous states
    - Example: vehicle control
  - Continuous controls on discrete states
    - Example: ratchet and pawl gears
  - Discrete controls on discrete states
    - Example: grid world
  - Discrete controls on continuous states
    - Example: water tank with binary in/out valve



Dynamic Equation

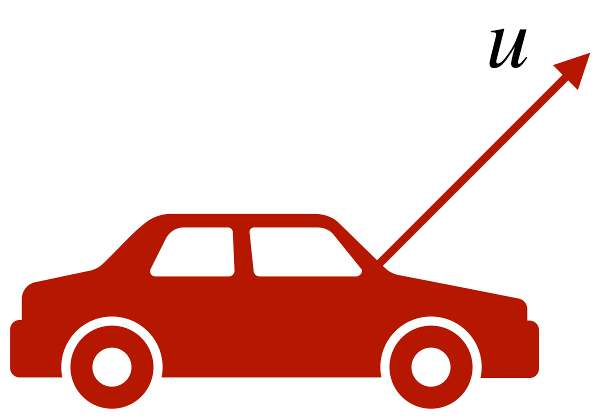
- A function describing the evolution of states under control inputs

	Deterministic Model	**Stochastic Model**
Continuous time (Ordinary differential equation)	$\dot{x} = f(x, u)$	$\dot{x} = f(x, u, w)$
Discrete time	$x_{k+1} = f(x_k, u_k)$	$x_{k+1} = f(x_k, u_k, w_k)$

Random variables

Example: Single Integrator Model

- State $x \in \mathbb{R}^2$: position
- Control $u \in \mathbb{R}^2$: velocity
- Dynamic equation
  - $\dot{x} = u$
  - $x_{k+1} = x_k + \Delta t \ u_k$

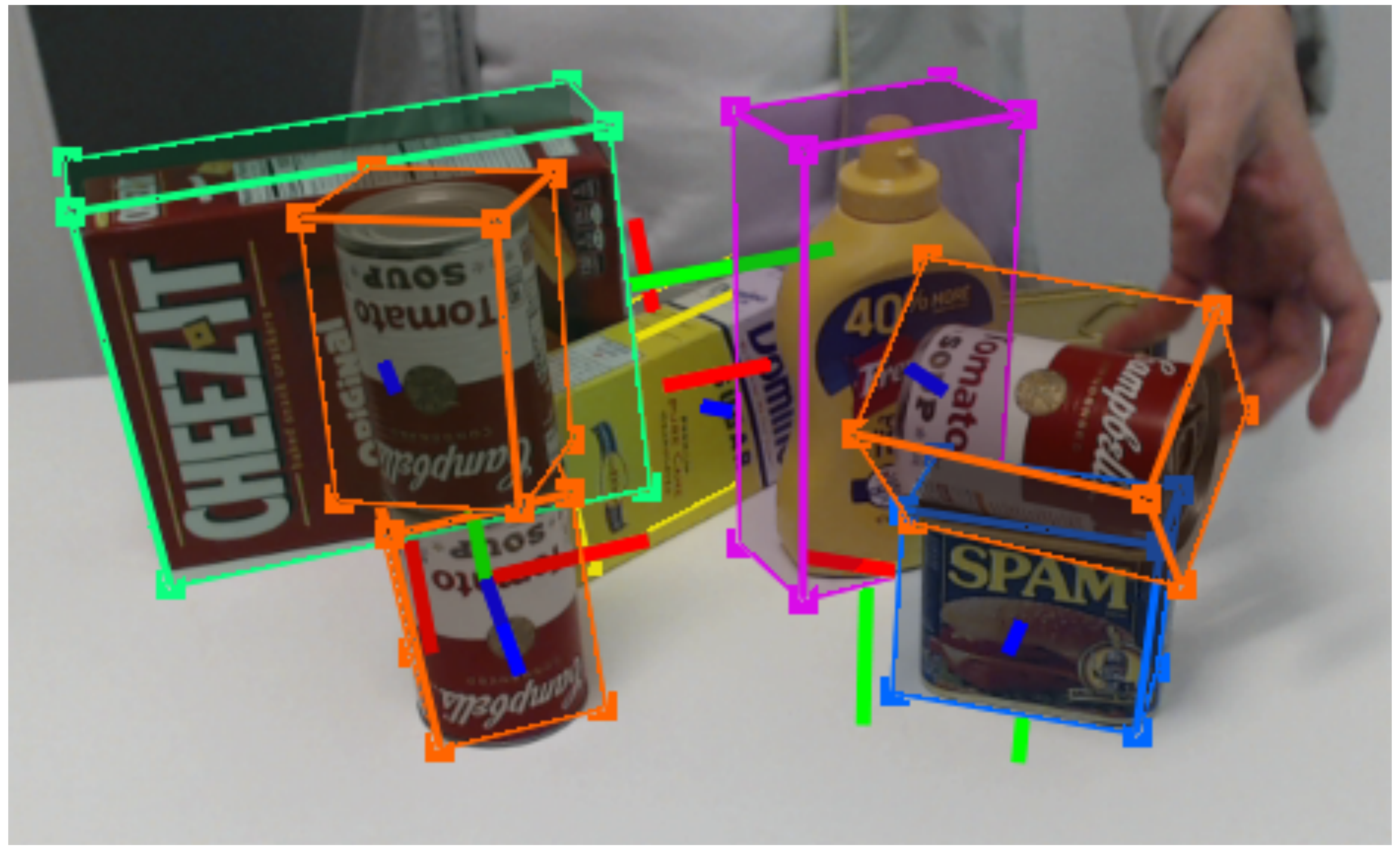


Control Decisions

- The control decision can be made based on
  - the current state (state-feedback control $u = g(x)$)
  - the measurement of the current state (measurement-feedback control)
  - historical states or their measurements (control with memory)

Observation Space

- In many cases, states are not directly measurable. They need to be inferred from measurements (y)
- Measurement function
  - Deterministic $y = h(x, u)$
  - Stochastic $y = h(x, u, v)$

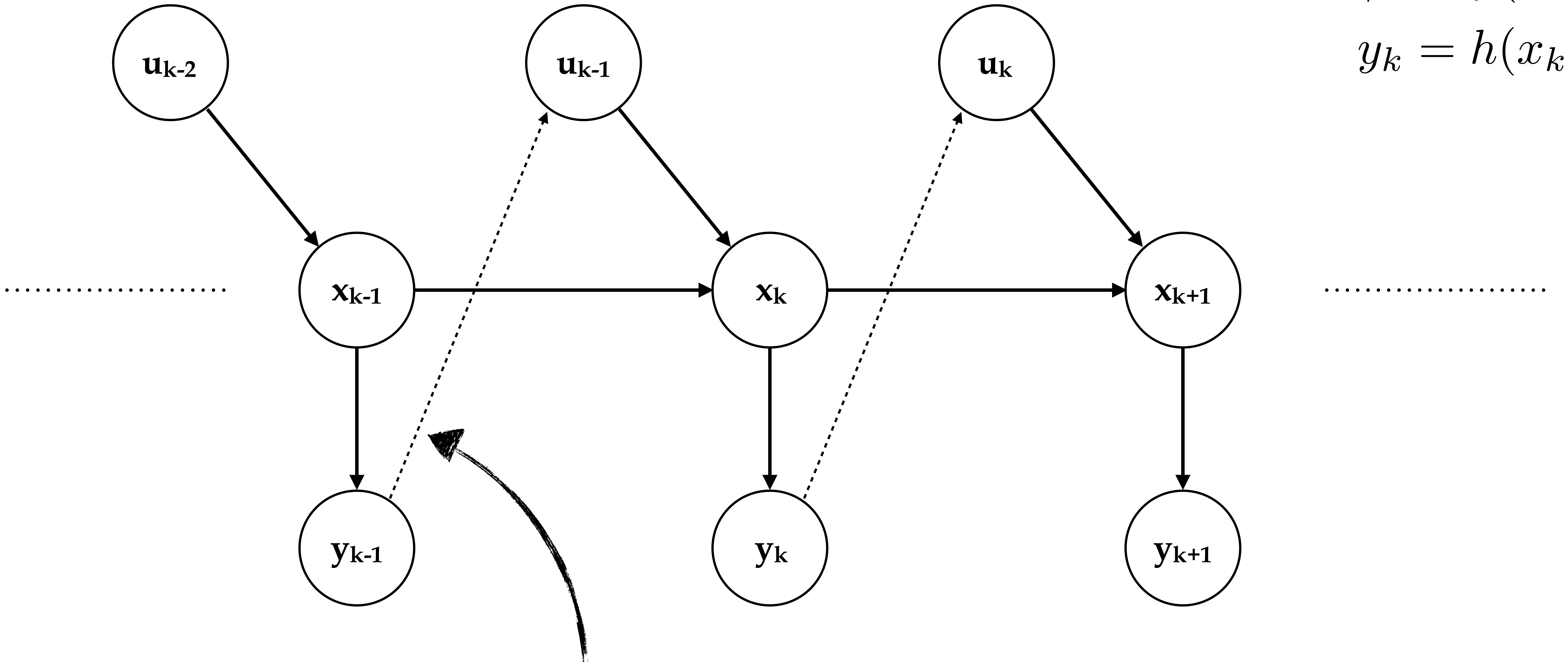


State Space Dynamic Model

	Deterministic Model	Stochastic Model
Continuous time	$\dot{x} = f(x, u)$ $y = h(x, u)$	$\dot{x} = f(x, u, w)$ $y = h(x, u, v)$
Discrete time	$x_{k+1} = f(x_k, u_k)$ $y_k = h(x_k, u_k)$	$x_{k+1} = f(x_k, u_k, w_k)$ $y_k = h(x_k, u_k, v_k)$

State Space Dynamic Model (Discrete Time)

$$x_{k+1} = f(x_k, u_k, w_k)$$
$$y_k = h(x_k, u_k, v_k)$$



Measurement-feedback control law

Rational Behaviors

- All control signals or actions are chosen to optimize certain objectives.
- Examples:
  - Stabilization at certain state (Regulation)
    - e.g., inverted pendulum
  - Trajectory tracking
    - e.g., industrial robot
  - Multi-objective control
    - e.g., autonomous driving

Objective Functions

- **Minimize cost J** or maximize reward R
- A cost function consists of
  - Run-time cost (incurred at every time instance)
    - Example: battery consumption cost for electronics
  - Terminal cost (if certain event happens or if at certain time)
    - Example: arriving at destination

Objective Functions

- Finite horizon
  - Terminate at certain time
    - Example: exams
  - Terminate at certain event
    - Example: goal-reaching problems
- Infinite horizon: no termination
  - Example: stabilization problems

Objective Functions

- Rational behavior is to always choose actions that minimize the current value of all discounted future costs.

- Example: Infinite horizon objective function

$$J = \sum_{k=0}^{\infty} \underbrace{\delta^k}_{\text{Discounting factor}} \underbrace{l_k(x_k, u_k, x_{k+1})}_{\text{Run-time cost}}$$

Controller Design Problem (Infinite Horizon)

- Given a state space **X**, a control space **U**, an observation space **Y**, an initial state, noise models in stochastic case
- We need to design a control law that minimizes the objective function

	Deterministic Model	**Stochastic Model**
Continuous time	$\dot{x} = f(x,u)$ $y = h(x,u)$ $J = \int_{t=0}^{\infty} \delta^t l_t(x,u,\dot{x})dt$	$\dot{x} = f(x,u,w)$ $y = h(x,u,v)$ $J = \int_{t=0}^{\infty} \delta^t l_t(x,u,\dot{x})dt$
Discrete time	$x_{k+1} = f(x_k,u_k)$ $y_k = h(x_k,u_k)$ $J = \sum_{k=0}^{\infty} \delta^k l_k(x_k,u_k,x_{k+1})$	$x_{k+1} = f(x_k,u_k,w_k)$ $y_k = h(x_k,u_k,v_k)$ $J = \sum_{k=0}^{\infty} \delta^k l_k(x_k,u_k,x_{k+1})$

Should be evaluated w.r.t. uncertainties

Markov Decision Process

- An MDP problem includes the following items:
  - A state space **X**, a control space **U**, an initial state
  - A transition function
    - $T(x_k, u_k, x_{k+1}) : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \to \mathbb{R}^+$
    - The probability of arriving at the particular state value at next step given the current state value and the current control action.
  - A reward function
    - $R(x_k, u_k, x_{k+1}) : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \to \mathbb{R}$
  - A discount factor δ

Markov Decision Process

- A MDP is equivalent to a **stochastic discrete-time state space model** with full state noise-free measurement:

MDP

- A state space **X**, a control space **U**, an initial state
- A transition function $T(x_k, u_k, x_{k+1}) : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \to \mathbb{R}^+$
- A reward function $R(x_k, u_k, x_{k+1}) : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \to \mathbb{R}$
- A discount factor δ

State Space Model

- A state space **X**, a control space **U**, an initial state
- A noise model of w_k
- A dynamic equation $x_{k+1} = f(x_k, u_k, w_k)$
- A cost function $J = \sum_{k=0}^{\infty} \delta^k l_k(x_k, u_k, x_{k+1})$

Partially Observed MDP

- A POMDP problem includes the following items:
  - A state space **X**, a control space **U**, an observation space **Y**, an initial state
  - A transition function
    - $T(x_k, u_k, x_{k+1}) : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \to \mathbb{R}^+$
  - An observation function
    - $O(y_{k+1}, x_{k+1}, u_k) : \mathcal{Y} \times \mathcal{X} \times \mathcal{U} \to \mathbb{R}^+$
    - The probability of obtaining certain measurement given the current state value and the last step's control action.
  - A reward function
    - $R(x_k, u_k, x_{k+1}) : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \to \mathbb{R}$
  - A discount factor δ

Partially Observed MDP

- A POMDP is equivalent to a **stochastic discrete-time state space model:**

POMDP

- A state space **X**, a control space **U**, an observation space **Y**, an initial state
- A transition function $T(x_k, u_k, x_{k+1}) : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \rightarrow \mathbb{R}^+$
- An observation function $O(y_{k+1}, x_{k+1}, u_k) : \mathcal{Y} \times \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}^+$
- A reward function $R(x_k, u_k, x_{k+1}) : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \rightarrow \mathbb{R}$
- A discount factor δ

State Space Model

- A state space **X**, a control space **U**, an observation space **Y**, an initial state
- Noise models of w_k and v_k
- A dynamic equation $x_{k+1} = f(x_k, u_k, w_k)$
- An observation equation $y_k = h(x_k, u_k, v_k)$
- A cost function
$$J = \sum_{k=0}^{\infty} \delta^k l_k(x_k, u_k, x_{k+1})$$

Next Time

- Kinematic and dynamic models